

Recommendation Systems for Personalized Content Discovery

Netflix RecSys Technical Report

1. Problem Understanding & Motivation

Recommendation systems power personalization across modern digital platforms such as Netflix, Amazon, and Spotify, directly impacting user engagement and retention. This project implements a collaborative filtering pipeline utilizing the historical Netflix Prize dataset (subsampling for computational efficiency to fit a 16 GB RAM environment). The primary goals are learning user preferences from interaction histories, predicting unseen ratings, and generating top-K personalized movie recommendations.

2. Dataset Characterization & EDA

The parsed raw dataset contains a total of 24,053,764 ratings from combined_data_1.txt. A representative sample of 1,000,000 interactions was utilized for training and validation.

Key properties observed:

- Star rating distribution is highly skewed towards 3 and 4 stars, with relatively few 1-star ratings.
- Sparsity was found to be approximately 98.8%, meaning users only rate a tiny fraction of total catalog movies.

3. Methodology & Models

Two core collaborative-filtering approaches were trained:

- Singular Value Decomposition (SVD): An explicit feedback matrix factorization model optimized using Stochastic Gradient Descent (SGD) via the Surprise library.
- Alternating Least Squares (ALS): An implicit feedback latent factor model implemented via the Implicit library, which models user preferences using rating values as confidence weights ($1 + \text{rating}$).

4. Experimental Evaluation

Using an out-of-sample 80-20 train-test split, we computed RMSE (rating accuracy) and MAP@10 (ranking quality). Relevance for MAP@10 was defined as a true rating ≥ 3.5 .

Model Performance Results:

- SVD RMSE: 0.9896
- ALS RMSE: 2.8183
- SVD MAP@10: 0.0000
- ALS MAP@10: 0.0057

Discussion: SVD achieved significantly lower RMSE because it is explicitly trained to minimize prediction error on rating scales. Conversely, ALS is an implicit feedback model that outputs user preference scores rather than exact star ratings, resulting in a higher RMSE.

5. Recommendation Examples & Insights

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For a sample user, the top-10 recommended movie IDs generated by the ALS model are:

30, 191, 28, 143, 312, 165, 45, 118, 9, 87

Observations:

- SVD handles cold-start users by utilizing the global average rating plus user/item biases.
- ALS generates high-coverage recommendations by mapping implicit user preference signals.

6. Future Improvements

- Hybrid Recommendation System: Incorporating movie release year and metadata.
- Neural Collaborative Filtering (NCF): Capturing non-linear interactions using deep networks.
- Hyperparameter tuning of learning rates and regularization variables.